

How Much Tensor Structure Should a Transformer FFN Have?

Routed Block-CP Feedforward Networks Between SwiGLU and Dense Tucker

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Abstract

[Abstract placeholder.]

1 Introduction

[Introduction placeholder.]

2 The routed tensor view of gated FFNs

Setup. At a fixed token position, let $x \in \mathbb{R}^d$ be the residual-stream vector entering the feedforward block. The SwiGLU FFN (Shazeer, 2020) computes

$$y = U^\top h, \quad h = (W^\top x) \odot \text{SiLU}(G^\top x), \quad (1)$$

with $W, G \in \mathbb{R}^{d \times m}$ and $U \in \mathbb{R}^{m \times d}$. Writing w_j, g_j for the j -th columns of W, G and u_j for the j -th column of $U^\top$, the j -th hidden coordinate is $h_j(x) = (w_j^\top x) \text{SiLU}(g_j^\top x)$.

2.1 SwiGLU is an exactly routed CP tensor field

Each SwiGLU channel multiplies two scalar projections of the residual stream. The SiLU nonlinearity factorizes exactly, $\text{SiLU}(z) = z \sigma(z)$ with $\sigma(z) = (1 + e^{-z})^{-1}$, so

$$h_j(x) = \underbrace{(w_j^\top x)(g_j^\top x)}_{\text{rank-one bilinear interaction}} \cdot \underbrace{\sigma(g_j^\top x)}_{\alpha_j(x) \in (0,1)}. \quad (2)$$

Defining the input-dependent interaction tensor $A(x) \in \mathbb{R}^{d \times d \times d}$ by $y_o = \sum_{a,b} A_{oab}(x) x_a x_b$, substituting (2) into (1) gives

$$A(x) = \sum_{j=1}^m \alpha_j(x) u_j \otimes w_j \otimes g_j, \quad \alpha_j(x) = \sigma(g_j^\top x). \quad (3)$$

This is a canonical polyadic (CP) decomposition (Kolda and Bader, 2009) of $A(x)$ with *shared, input-independent factors* and *input-dependent routing weights* $\alpha_j(x)$. The decomposition is exact and holds pointwise in x ; no approximation is involved. We call $u_j \otimes w_j \otimes g_j$ the j -th *interaction*

atom (what interaction is available) and $\alpha_j(x)$ its *route* (when it is used). The bilinear $\alpha \equiv 1$ limit of this picture is the bilinear MLP analyzed by [Pearce et al. \(2025\)](#); the routed form keeps the gate inside the structural description rather than discarding it.

Viewed as a Tucker decomposition with latent dimensions $r = s = m$ and factor matrices $(P, Q, R) = (W, G, U^\top)$, the CP form (3) corresponds to a *superdiagonal core*: $C_{oij} = \delta_{oi}\delta_{ij}$. This restriction cuts both ways. It atomizes the layer — every hidden index is one (atom, route) pair, and CP decompositions are essentially unique under mild conditions ([Kruskal, 1977](#)), so the atoms are identifiable objects. But it also means one route controls exactly one rank-one interaction: feature w_i never interacts with gate g_j for $i \neq j$.

2.2 Dense Tucker and the per-route interaction matrix

The unconstrained relaxation drops the superdiagonal restriction. A *Tucker-core FFN* with latent projections $P, Q \in \mathbb{R}^{d \times r}$, output basis $R \in \mathbb{R}^{d \times s}$, and learned core $C \in \mathbb{R}^{s \times r \times r}$ computes $p = P^\top x$, $q = Q^\top x$,

$$z_o = \sum_{i,j=1}^r C_{oij} p_i \text{SiLU}(q_j), \quad y = Rz. \quad (4)$$

Grouping by gate index exposes the structure this paper is about:

$$y(x) = \sum_{j=1}^r V_j p \text{SiLU}(q_j), \quad V_j := R C^{(j)} \in \mathbb{R}^{d \times r}, \quad C_{oi}^{(j)} = C_{oij}. \quad (5)$$

Each latent gate q_j routes a *matrix* V_j of output-by-main interactions. SwiGLU is the corner case $\text{rank } V_j = 1$ for all j ; a generic Tucker core has $\text{rank } V_j = \min(r, s)$.

Two properties make the dense core suspicious as an interpretability target. First, *gauge freedom*: for any invertible $M \in \mathbb{R}^{r \times r}$, replacing $P \mapsto PM^{-\top}$ and $C^{(j)} \mapsto C^{(j)} M^\top$ for all j leaves the function unchanged, so individual latent directions p_i and individual core entries carry no intrinsic meaning — only the gate directions $Q_{:j}$ and the subspaces $\text{range}(V_j)$ are well defined. Second, the core costs sr^2 parameters; at the configurations we study it absorbs over 90% of the block’s budget.

The aligned-width theorem, reinterpreted. Prior work on this codebase proved an exact separation: if $[P \ Q]$ has full column rank, an *aligned* SwiGLU (units restricted to main vectors in $\text{span}(P)$ and gate vectors from the columns of Q) that exactly realizes (5) must have width $m \geq \sum_j \text{rank } V_j$, and a rank decomposition of each V_j attains the bound ([Appendix A](#)). Generically this is $r \min(r, s)$, i.e. k^2 in the balanced case — the result that originally motivated dense Tucker as “ $k \times$ cheaper than SwiGLU.” The reading we adopt here is different: *the theorem says the operative variable is the per-route rank* $\text{rank } V_j$, not the presence of a dense core. The constructive half of the proof — realize each V_j as a sum of $\text{rank } V_j$ rank-one terms sharing the same gate — is itself an architecture. That architecture is the subject of this paper.

2.3 LL1/block-CP: one route, one rank- L block

A third-order tensor admits a *decomposition in multilinear rank- $(L, L, 1)$ terms* (LL1; a structured block-term decomposition) if it can be written $T = \sum_{b=1}^B (A_b B_b^\top) \otimes c_b$: each term is rank- L in two modes and rank-one in the third ([De Lathauwer, 2008](#); [Vervliet et al., 2016](#)). Taking the *gate mode* as the rank-one mode yields the FFN we study. The **LL1 FFN** with B blocks of rank L has, per

block, a gate direction $g_b \in \mathbb{R}^d$, a main factor $A_b \in \mathbb{R}^{d \times L}$, and an output factor $U_b \in \mathbb{R}^{d \times L}$:

$$y(x) = \sum_{b=1}^B U_b (A_b^\top x) \text{SiLU}(g_b^\top x). \quad (6)$$

Expanding into atoms, $y(x) = \sum_b \sum_{l=1}^L u_{b,l} (a_{b,l}^\top x) \text{SiLU}(g_b^\top x)$: *grouped CP*, L rank-one atoms sharing one route. The interaction tensor is the routed LL1 tensor $A(x) = \sum_b \alpha_b(x) (U_b A_b^\top) \otimes g_b$, and the per-route matrix is $V_b = U_b A_b^\top$ with $\text{rank } V_b \leq L$ *by construction*. The aligned-width theorem’s control variable has become a hyperparameter.

The family nests both endpoints exactly. At $L = 1$, $B = m$, (6) is SwiGLU (verified to machine precision in our implementation); a single block with $L = r$ is a (one-gate) dense slice; and an LL1 FFN equals a Tucker FFN whose core is block-superdiagonal. LL1 decompositions also retain essential-uniqueness guarantees under mild conditions (De Lathauwer, 2008, 2011), unlike the gauge-free dense core.

What the parameter budget buys. The comparison that matters is at matched parameters:

$$N_{\text{SwiGLU}} = 3dm, \quad N_{\text{LL1}} = dB(2L + 1), \quad N_{\text{Tucker}} = d(2r + s) + sr^2. \quad (7)$$

At a fixed budget N , SwiGLU affords $m = N/3d$ atoms, each with a private route; LL1 affords $BL = \frac{3L}{2L+1} \cdot \frac{N}{3d}$ atoms — up to 1.5× more — but only $B = \frac{3}{2L+1} \cdot \frac{N}{3d}$ routes. *LL1 trades routing diversity for interaction atoms*. An equivalent statement: an LL1(B, L) block is exactly a width- BL SwiGLU whose gate vectors are tied in groups of L . Which side of the trade wins is an empirical question about what trained FFNs actually need, and it is the question our experiments are designed to answer. A prior observation sharpens the stakes: in a dense Tucker LM trained from scratch, the learned per-gate stable rank $\|V_j\|_F^2 / \|V_j\|_{\text{op}}^2$ concentrates around 4 — far below the generic value r , far above the SwiGLU value 1. The trained dense core already behaves like an LL1 model; LL1 makes that behavior an explicit, cheaper, and identifiable parameterization.

3 Experiments

3.1 Synthetic teacher–student recovery

[exp18 placeholder]

3.2 Matched-budget language modeling

[exp11/sprint_lm placeholder]

3.3 Interpretability proxies

[exp19/exp22 placeholder]

3.4 Circuit-level pilot: induction

[exp20 placeholder]

4 Discussion

[placeholder: what is established / plausible / open; limitations; next experiment.]

5 Related work

[placeholder]

References

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A The aligned-width theorem

This appendix restates, with proof, the separation theorem from the earlier draft accompanying this codebase, and records its constructive half as the LL1 architecture. Throughout, $\phi = \text{SiLU}$; the only properties used are $\phi(0) = 0$ and $\phi(1) \neq 0$.

Lemma 1 (SwiGLU recovery). *Setting $r = s = m$, $P = W$, $Q = G$, $R = U^\top$, and $C_{oi} = \delta_{oi}\delta_{ij}$ in (4) reduces the Tucker-core FFN to the SwiGLU map (1).*

Proof. With the superdiagonal core, (4) collapses to $z_o = p_o \phi(q_o)$, so $z = (P^\top x) \odot \phi(Q^\top x)$, which is h from (1); then $y = Rz = U^\top h$. $\square$

Definition 1 (Aligned SwiGLU representation). *Given latent matrices $P, Q \in \mathbb{R}^{d \times r}$, an aligned width- m SwiGLU representation is a map*

$$\hat{y}(x) = \sum_{j=1}^r \sum_{\nu=1}^{k_j} u_{j\nu} (a_{j\nu}^\top p) \phi(q_j), \quad \sum_j k_j = m,$$

with $u_{j\nu} \in \mathbb{R}^d$, $a_{j\nu} \in \mathbb{R}^r$, $p = P^\top x$, $q = Q^\top x$. Each unit uses a main vector in $\text{span}(P)$ and one of the gate vectors $Q_{:j}$.

Theorem 1 (Diagonal bottleneck). *Assume $[P \ Q] \in \mathbb{R}^{d \times 2r}$ has full column rank. The minimum width of an aligned SwiGLU representation exactly realizing a Tucker-core FFN with per-gate coefficient matrices $\{V_j\}_{j=1}^r$ is $m_{\min} = \sum_{j=1}^r \text{rank } V_j$.*

Proof. Lower bound. By the rank assumption, $x \mapsto (p, q)$ is surjective onto $\mathbb{R}^r \times \mathbb{R}^r$, so an exact representation forces $\sum_j (V_j - \hat{V}_j) p \phi(q_j) = 0$ for all (p, q) , where $\hat{V}_j = \sum_\nu u_{j\nu} a_{j\nu}^\top$ has rank at most k_j . Fix j_0 and set $q_j = \delta_{jj_0}$; since $\phi(0) = 0$ and $\phi(1) \neq 0$, only the j_0 term survives and $V_{j_0} = \hat{V}_{j_0}$. Hence $k_j \geq \text{rank } V_j$ for every j , and $m = \sum_j k_j \geq \sum_j \text{rank } V_j$.

Tightness. Let $V_j = \sum_{\nu=1}^{\rho_j} u_{j\nu} a_{j\nu}^\top$ be any rank decomposition, $\rho_j = \text{rank } V_j$. Then $V_j p \phi(q_j) = \sum_\nu u_{j\nu} ((Pa_{j\nu})^\top x) \phi(Q_{:,j}^\top x)$ is a sum of ρ_j aligned units sharing gate $Q_{:,j}$. $\square$

Corollary 1 (Generic separation). *If R has full column rank and each slice $C^{(j)}$ is generic, then $\text{rank } V_j = \min(r, s)$ for all j and $m_{\min} = r \min(r, s)$; in the balanced case $r = s = k$, $m_{\min} = k^2$.*

Remark 1 (The constructive half is LL1). *The tightness construction — ρ_j rank-one units sharing the gate $Q_{:,j}$ — is precisely one block of the LL1 FFN (6) with $L = \rho_j$. Theorem 1 therefore does not say “dense cores beat SwiGLU”; it says the cost of simulating a routed tensor map is the sum of its per-route ranks, and an architecture that fixes those ranks directly realizes the map at exactly that cost.*

B Experimental details

[hyperparameters, configs, hardware.]